Recovery and always-on supply



File: recovery.kicad_sch

Processor PMIC



File: pmic.kicad_sch

DDR BUCK3



File: ddr-power.kicad_sch

CPU dynamic voltage supply



File: cpu-power.kicad_sch

Main 3.3V



File: main3v3.kicad_sch

Main 5V control



File: main5v.kicad_sch

Core local bypass



File: core-power.kicad_sch

Boot storage and clock



File: boot.kicad_sch

RK3566 processor



File: processor.kicad_sch

USB analog filters



File: phy-power.kicad_sch

Micron RAM



File: ram.kicad_sch

Host recovery interfaces



File: host-interface.kicad_sch

Peripheral supply and feedback



File: peripheral-power.kicad_sch

2.5GbE controller



File: nic.kicad_sch

Ethernet power sequencing



File: nic-power.kicad_sch

PCIe and power-cycle interfaces



File: nic-interface.kicad_sch

Ethernet local bypass



File: nic-bypass.kicad_sch

PoE active input bridge



File: poe-input.kicad_sch

PoE PD / Class 6



File: poe-pd.kicad_sch

Isolated 12V flyback



File: poe-flyback.kicad_sch

Isolated PoE allocation



File: poe-status.kicad_sch

Ethernet cable ingress



File: ethernet-ingress.kicad_sch

Modem independent power



File: modem-power.kicad_sch

Modem and SIM sockets



File: modem-socket.kicad_sch

Modem USB isolation



File: modem-interface.kicad_sch

Modem local bypass



File: modem-bypass.kicad_sch

Complete appliance connectivity captured; engineering review.
Four module assembly profiles require current controlled guides.
Footprint release, layout, isolation, thermal, USB/PCIe and recovery
qualification remain required before manufacturing.

Sheet: /
File: appliance.kicad_sch

Title: Cellular WAN appliance — engineering capture

Size: A1	Date: 2026-10-07	Rev:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 1/30

SCOPE OF THIS SCHEMATIC MILESTONE

Implemented: MCU, independent watchdog/brownout, SWD, manual reset, open-collector control outputs and default-off enables.
Not implemented: packet processor, PoE/isolation, modem socket/SIM, high-current regulators or Ethernet. NOT a complete WAN appliance.
J101 is a 3.3V rail test header; J102 is an integration boundary, not a modem socket. No PCB layout exists.

Supervisor

MCU_MODEM_POWER_EN
MCU_ETH_POWER_EN
SERVICE_TX
SERVICE_RX
MODEM_RESET_ASSERT
MODEM_OFF_ASSERT
ETH_RESET_ASSERT
MCU_HEALTH_IN
POWER_FAULT_N

MODEM_POWER_EN
ETH_POWER_EN
SERVICE_TX
SERVICE_RX
MODEM_RESET_ASSERT
MODEM_OFF_ASSERT
ETH_RESET_ASSERT
HEALTH_IN
POWER_FAULT_N

File: supervisor.kicad_sch

Recovery outputs

MODEM_RESET_ASSERT
MODEM_OFF_ASSERT
ETH_RESET_ASSERT
MODEM_RESET_N
MODEM_OFF_N
ETH_PERST_N
MCU_MODEM_POWER_EN
MCU_ETH_POWER_EN

MODEM_RESET_ASSERT
MODEM_OFF_ASSERT
ETH_RESET_ASSERT
MODEM_RESET_N
MODEM_OFF_N
ETH_RESET_N
MODEM_POWER_EN
ETH_POWER_EN

File: controls.kicad_sch

Always-on supply

AON_POWER_GOOD
POWER_FAULT_N

File: always-on.kicad_sch

+3V3
1
2
GND
J101
3.3V RAIL TEST

3.3V rail test; do not back-power.
No PoE or unisolated input here.

J102
CONTROL BOUNDARY

MCU_MODEM_POWER_EN	1
MCU_ETH_POWER_EN	2
MODEM_RESET_N	3
MODEM_OFF_N	4
ETH_PERST_N	5
SERVICE_TX	6
SERVICE_RX	7
MCU_HEALTH_IN	8
POWER_FAULT_N	9
GND	10

Service UART / health / fault inputs:
3.3V only. Add powered-off-safe translation
when target voltage requires it.
Service UART must not back-power target.

No MCU modem-session interface is implemented.

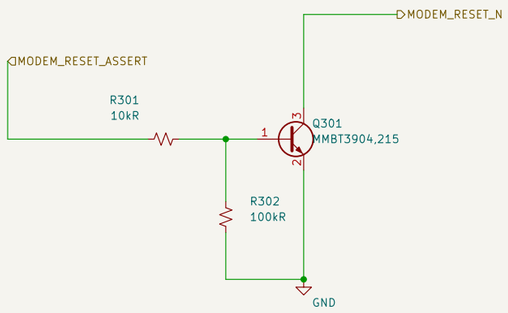
REVIEW ONLY — independent recovery core
sierra-to-usb

Sheet: /Recovery and always-on supply/
File: recovery.kicad_sch

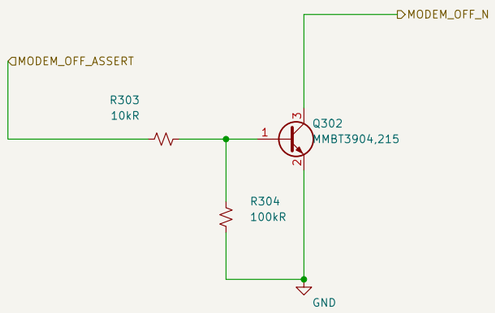
Title: Independent recovery core — architecture gate open

Size: A3	Date: 2026-09-28	**Rev: 0.1**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 2/30

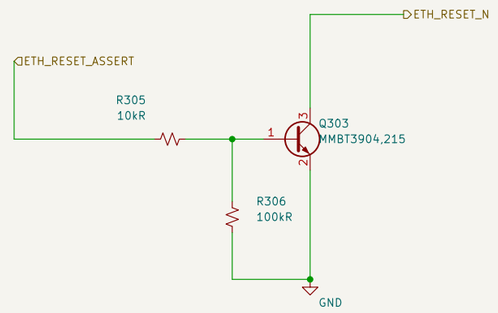
OPEN-COLLECTOR RESET / OFF — NO EXTERNAL PULL-UP TO ALWAYS-ON RAIL



HIGH input asserts LOW output.
Target pull-up / timing is profile-specific.

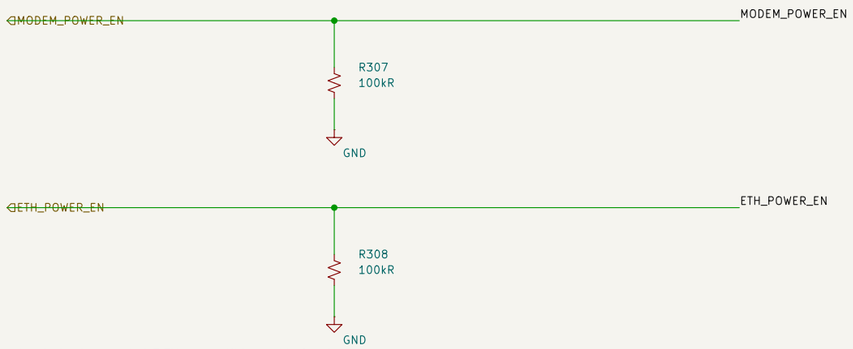


HIGH input asserts LOW output.
Target pull-up / timing is profile-specific.



HIGH input asserts LOW output.
Target pull-up / timing is profile-specific.

POWER ENABLES — DEFAULT OFF ON MCU RESET



Connect only to documented 3.3V-compatible enable inputs.
The regulators / eFuses are NOT captured in this milestone.

INTEGRATION LIMITS

Reset outputs tolerate target pull-ups without sourcing voltage into an unpowered target. Maximum target sink current allocation: 1mA; verify low threshold with exact load.

MODEM_OFF_N connects to FULL_CARD_POWER_OFF# only with a switched target-rail pull-up. Without that pull-up, the module remains off. Do not drive from an always-on rail.

This circuit does not implement modem power switching, PoE, USB forwarding, Ethernet or SIM. See architecture gate.

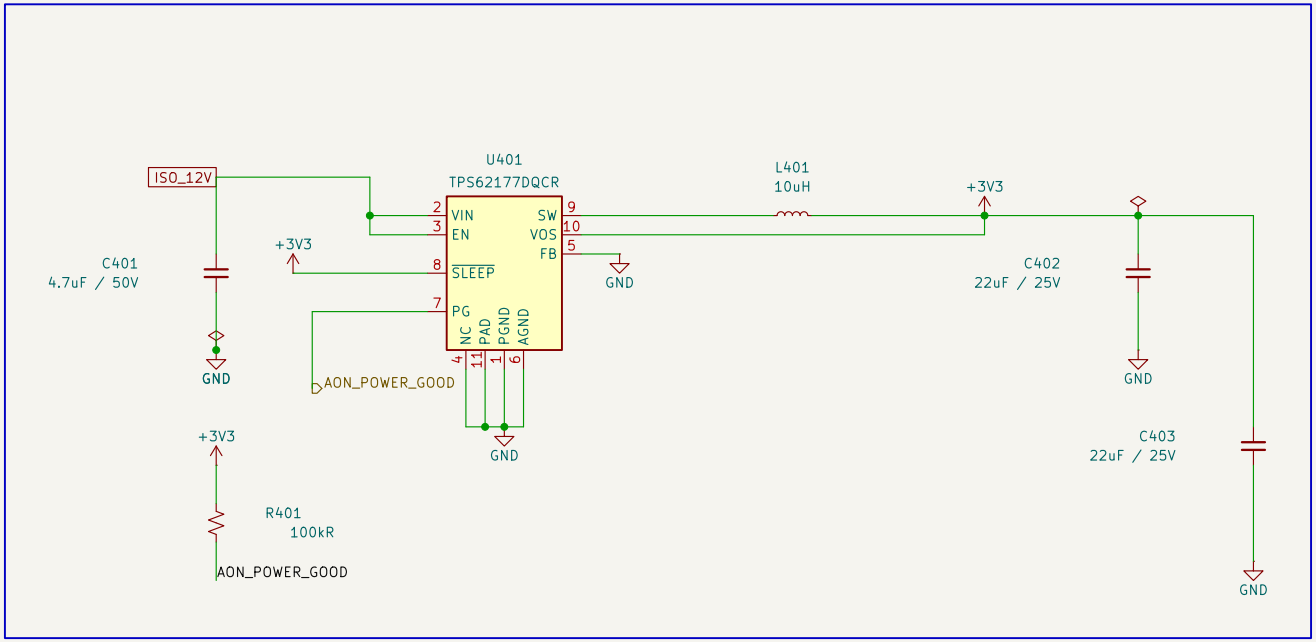
REVIEW ONLY — independent recovery core
sierra-to-usb

Sheet: /Recovery and always-on supply/Recovery outputs/
File: controls.kicad_sch

Title: Recovery outputs — safe control interfaces

Size: A3	Date: 2026-09-28	**Rev: 0.1**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 3/30

Independent supervisor supply



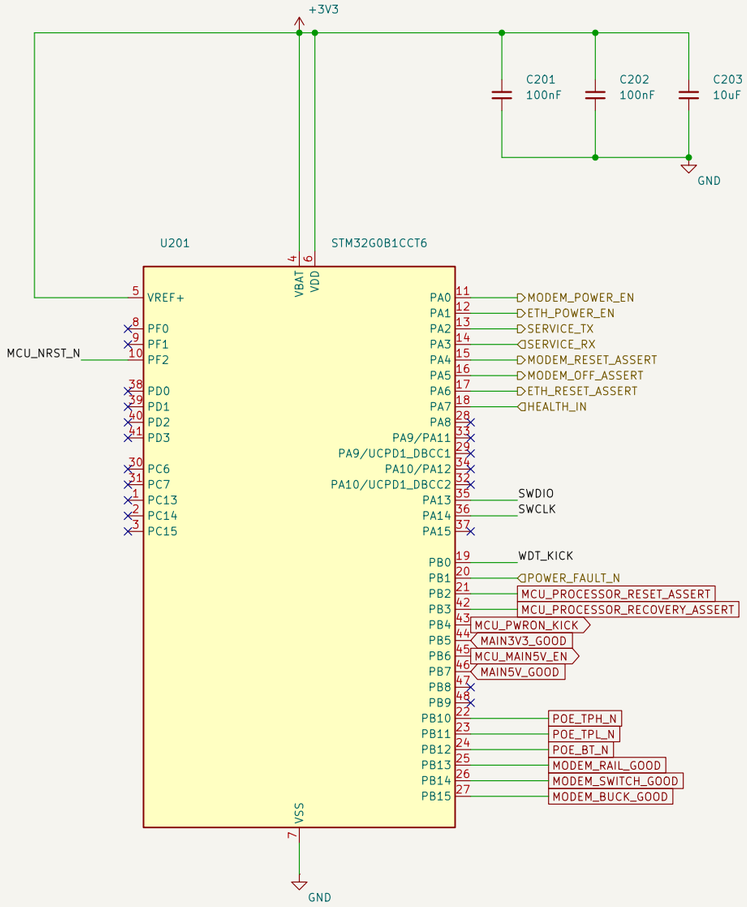
DRAFT: isolated 12V input; 3.3V / 0.5A output. No connection to raw PoE.
EN tied to input; SLEEP tied to 3.3V for normal operation.
Always on while the isolated input is present; downstream cuts must not gate this rail.

TI SLVSB35C, Figure 8 / Tables 2–3. 10uH + 2x22uF nominal.
Capacitor DC bias, complete rail load and layout review remain open.
Route VOS as Kelvin sense after L401; join AGND / PGND / exposed pad locally.

Sheet: /Recovery and always–on supply/Always–on supply/		
File: always–on.kicad_sch		
Title: Always–on supervisor supply		
Size: A4	Date: 2026–10–03	**Rev: 0.1–draft**
KiCad E.D.A. 10.99.0–4413–g83b5faf3d5		Id: 4/30

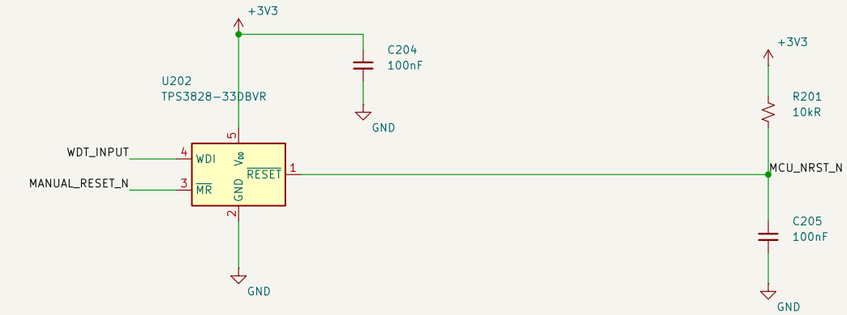
SUPERVISOR MCU — CONTROL ONLY

100nF at VDD/VREF; 10uF local bulk.
No external clock required for UART/GPIO supervision.



PF2 / pin 10 MUST remain NRST via option bytes.
Internal BOR and IWDG enabled by protected boot firmware.
PA2/PA3: USART2 AF1; PA13/PA14: SWD.
UART is service only; no modem AT transport selected.
Two signed app slots + protected loader planned; not implemented.

EXTERNAL WATCHDOG + BROWNOUT



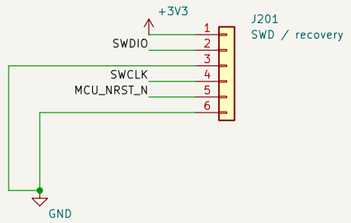
1.6s nominal watchdog; 2.93V threshold; 200ms nominal reset.
Open-drain RESET directly shares MCU NRST with SWD.
Verify NRST low-voltage margin across temperature before release.



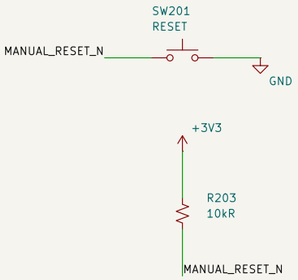


Fit RUN jumper for unattended use.
Remove only for physical SWD service.
WDI floats only with jumper removed.

PROGRAMMING / MANUAL RECOVERY



VTref only; debugger must not power the target.
Leave debug unlocked until signed-update recovery is proven.
No USB data path through this MCU.



REVIEW ONLY — independent recovery core
sierra-to-usb

Sheet: /Recovery and always-on supply/Supervisor/
File: supervisor.kicad_sch

Title: Supervisor — MCU, reset, watchdog and SWD

Size: A3	Date: 2026-09-28	**Rev: 0.1**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 5/30

PMIC — buck / analog / IO rails

U501A
RK809-5

5V_MAIN
3V3_MAIN
VDD_DDR_1V1
GPU_FB
LOGIC_FB
DDR_FB
VCC_1V8
PMIC_SLEEP
PMIC_PWRON_N
PMIC_VDC
PMIC_SW2
PMIC_SW1
PMIC_SW3
PMIC_SW5
UNUSED_SWITCH1
UNUSED_SWITCH2
PMIC_SW4
C524
22uF / 25V
C504
22uF / 25V
C538
1uF / 25V

BUCK1: 0.9V startup / 2.5A rated: 3 x 22uF output bank

L501
0.47uH
PMIC_SW1
C501
22uF / 25V
C502
22uF / 25V
C503
22uF / 25V
VDD_LOGIC
C533
22uF / 25V

NPU BUCK4: 1.5A; startup OFF. Program 0.9V before enable. Local load bypass captured; qualification pending.

L504
0.47uH
PMIC_SW4
VDD_NPU
C541
22uF / 25V
C542
10uF / 25V
R512
100R
NPU_FB

LDO2: 0.9V / 400mA; load budget pending

VDDA_0V9
C520
1uF / 25V
C521
1uF / 25V

U501B
RK809-5

5V_MAIN
3V3_MAIN
C516
22uF / 25V
C519
22uF / 25V
C508
22uF / 25V

BUCK2: 0.9V startup / 2.5A rated: 3 x 22uF output bank

L502
0.47uH
PMIC_SW2
VDD_GPU
C505
22uF / 25V
C506
22uF / 25V
C507
22uF / 25V

LDO3: 0.9V / 100mA high PSRR; load budget pending

VDDA0V9_PMU
C522
1uF / 25V
C523
1uF / 25V

VCCA1V8_PMU
VCCIO_SD
UNUSED_LDO9
VDDA0V9_PMU
VDDA_0V9
UNUSED_LDO1
UNUSED_LDO4
VCC3V3_PMU
C517
1uF / 25V
C518
1uF / 25V

U501C
RK809-5

C514
1uF / 25V
C512
22uF / 25V

BUCK5: 1.8V startup / 2.5A rated: 3 x 22uF output bank

L505
0.47uH
PMIC_SW5
VCC_1V8
C509
22uF / 25V
C510
22uF / 25V
C511
22uF / 25V

LDO7: 1.8V / slot 2: load budget pending

VCCA1V8_PMU
C525
1uF / 25V
C526
1uF / 25V

VCC3V3_PMU
R501
10kR
C513
100nF / 25V
C515
1uF / 25V

U501D
RK809-5

5V_MAIN
UNUSED_CODEC_LRCLK
UNUSED_CODEC_BCLK
PMIC_INTERNAL_1V8A
PMIC_INTERNAL_1V8D
C535
1uF / 25V
C536
1uF / 25V

LDO8: 1.8V / slot 2: load budget pending

VCCA_1V8
C527
1uF / 25V
C528
1uF / 25V

C537
10uF / 50V

R506
100kR
R507
100kR
C531
100nF / 25V
PMIC_VDC

LDO5: 3.3V startup / slot 4; SD IO adjustable; load budget pending

VCCIO_SD
C529
1uF / 25V
C530
1uF / 25V

R502
2.2kR
R503
2.2kR
R504
10kR
PMIC_I2C_SCL
PMIC_I2C_SDA
PMIC_INT_N

R505
10kR
PMIC_SLEEP

I2C0: 3.3V PMUIO2: 2.2kR pull-ups. Capacitance and firmware speed review pending.

R510
10kR
UNUSED_CODEC_LRCLK
R511
10kR
UNUSED_CODEC_BCLK

C532
100nF / 25V
R508
10kR
MCU_PWRON_MCK
Q501
MMBT3904,215
R509
100kR

PWRON: internal 35kR pull-up to 5V RTC. Open-drain control required.

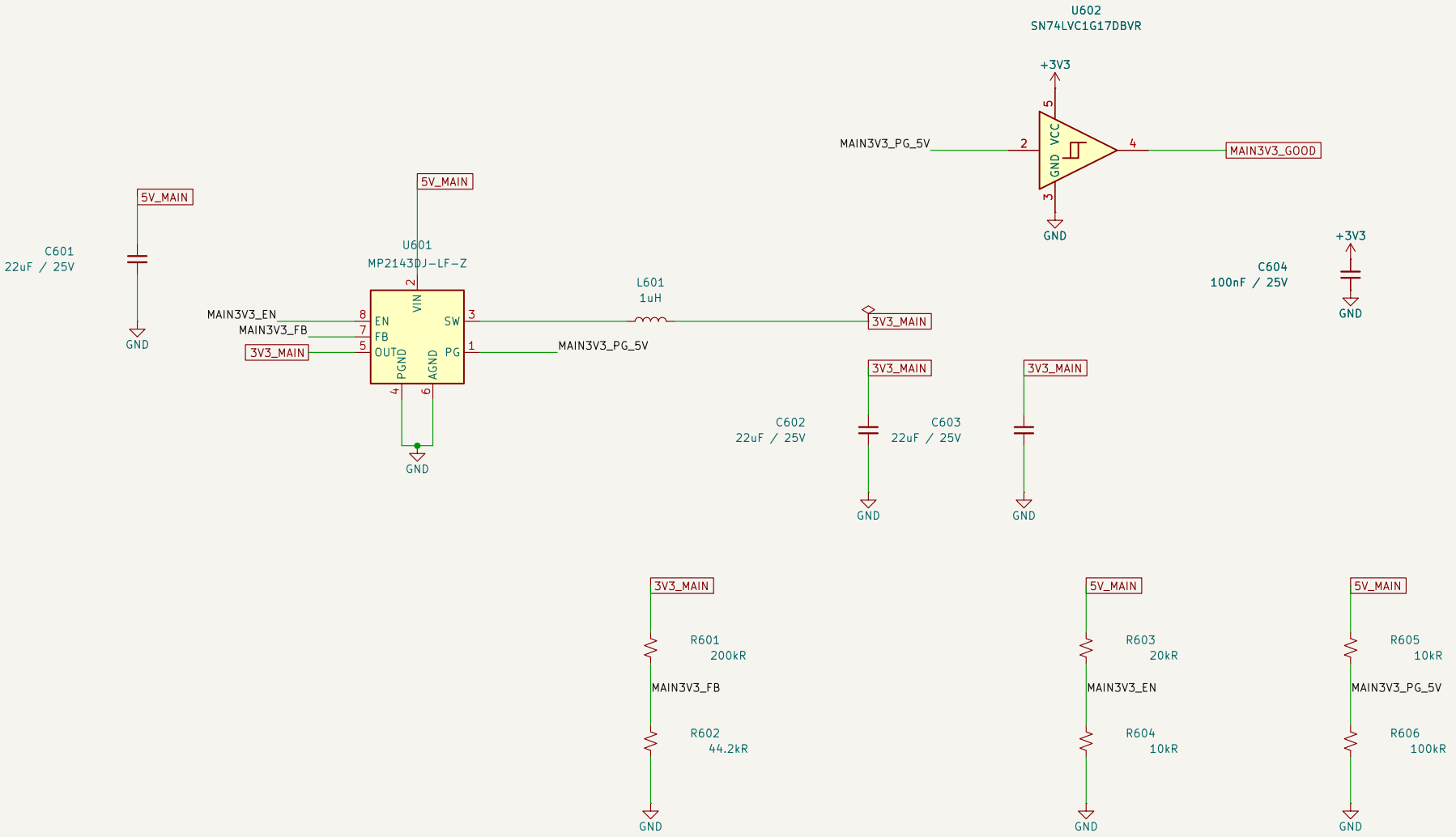
VDC: 100kR / 100kR, 100nF; nominal 2.5V at main 5V. Startup timing pending.

CPU EXT_EN connected. BATDIV/SNSP/SNSN grounded in no-battery configuration.
Unused LDO1/4/9 and SWOUT1/2 terminate in output bypasses; disable unused rails in firmware.
Internal RC oscillator per RK809 4.4.6; network time required. RTC accuracy not provided.
Remote feedback, load budgets, rail sequence and brownout recovery require bench qualification.

UNUSED_LDO1
C545
1uF
UNUSED_LDO4
C546
1uF
UNUSED_LDO9
C547
1uF
UNUSED_SWITCH1
C543
10uF
UNUSED_SWITCH2
C544
10uF

GND

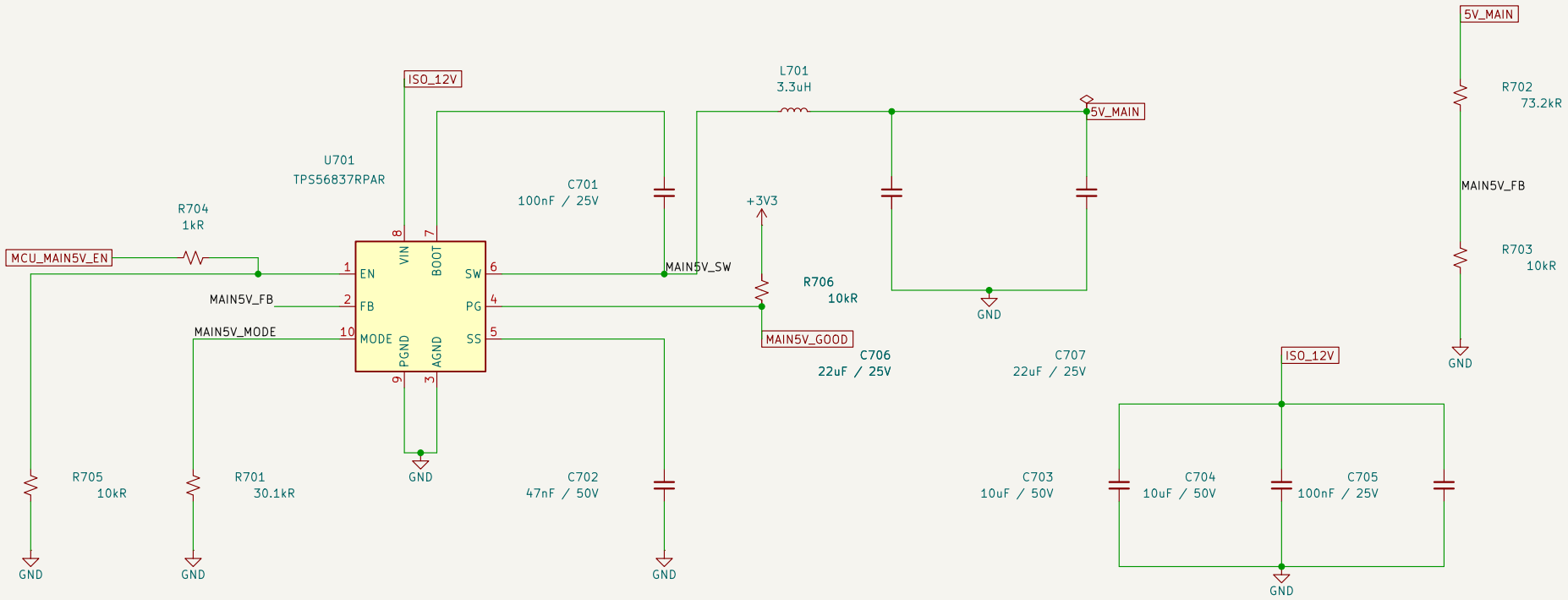
Sheet: /Processor PMIC/	
File: pmic.kicad_sch	
Title: RK809-5 power and control	
Size: A2	Date: 2026-10-07
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5	**Rev: 0.1-draft**
	Id: 6/30



Main sequencing and PG translation captured; capacitance and thermal margins unqualified.

Sheet: /Main 3.3V/	
File: main3v3.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5	Id: 7/30

MAIN 5 V CONTROL — ENGINEERING CAPTURE



MODE: 500 kHz; high valley current limit option. Feedback: 4.992 V nominal.

DRAFT: hot current margin, startup, input/output capacitance and loop stability unqualified.

Power path captured. Regulator footprint, stability and thermal release remain pending.

Sheet: /Main 5V control/ File: main5v.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 8/30

PROCESSOR: 565 balls captured. CPU/eMMC/PCIe/USB and PMIC controls connected.
Unused GPIO/analog signal balls are NC; disabled peripherals must remain disabled in firmware.
TVSS and XOUT24M float per reference; 1.8V active oscillator feeds XIN24M.
1.8V flash strap, debug strap and thermal-shutdown termination are on host-interface sheet.

U801A
RK3566

U801B
RK3566

U801C
RK3566

U801D
RK3566

U801E
RK3566

U801F
RK3566

U801G
RK3566

U801H
RK3566

R802
10kR

Q801
MMBT3904.215

R803
100kR

R804
10kR

Q802
MMBT3904.215

R805
100kR

R801
10kR

SW802
PROCESSOR / PMIC RESET

SW801
PROCESSOR RECOVERY

C801
1uF

C802
100nF

C803
1uF

C804
100nF

C805
1uF

C806
100nF

C807
1uF

C808
100nF

C809
1uF

C810
100nF

C811
100nF

C812
100nF

C815
100nF

C816
100nF

C817
100nF

C818
100nF

C819
100nF

C820
1nF

C821
100nF

C822
22uF

C823
10uF

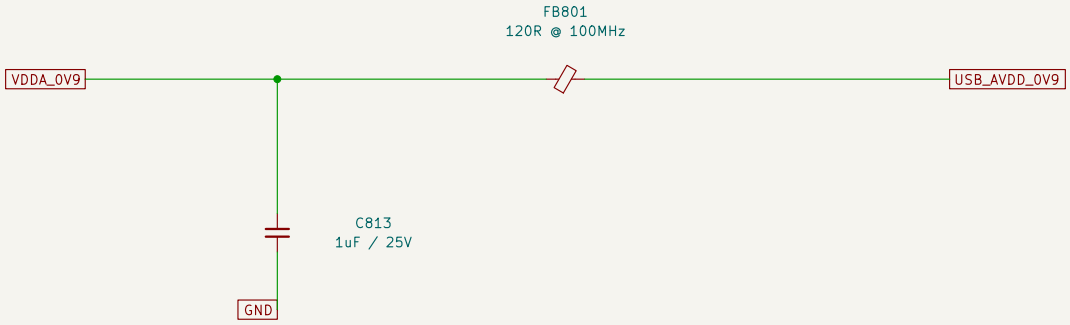
C824
22uF

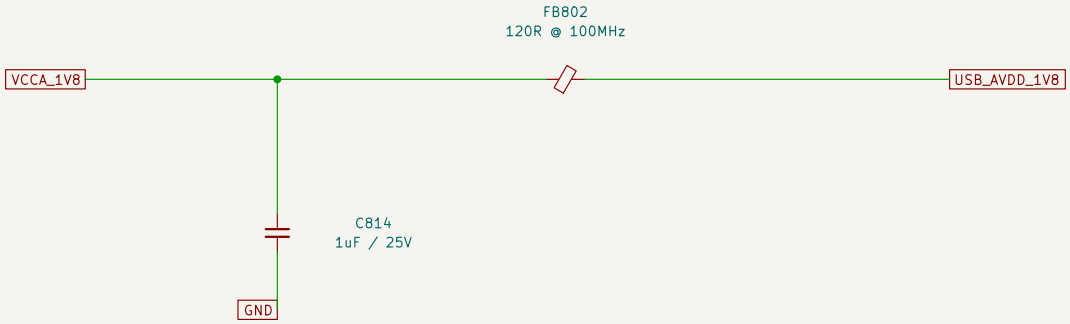
Sheet: /RK3566 processor/
File: processor.kicad_sch

Title: RK3566 processor — interface and power capture

Size: A2	Date: 2026-10-07	Rev:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 9/30

USB ANALOG FILTERS: local pin bypass and processor loads are captured.
45mOhm maximum DCR differs from 30mOhm reference example.
Rail headroom, thermal limits and bead/capacitor damping remain to verify.





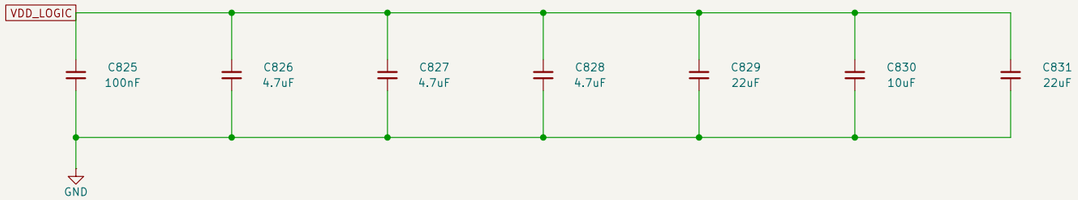
Sheet: /USB analog filters/
File: phy-power.kicad_sch

Title: USB PHY analog filters — engineering capture

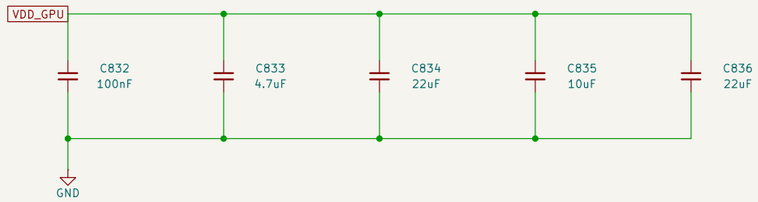
Size: A4	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 10/30

DRAFT CORE BYPASS: NanoPi R3S LTS 2411 sheets 5/6 nominal values.
Effective capacitance, layout feasibility, PDN impedance and regulator stability remain to verify.

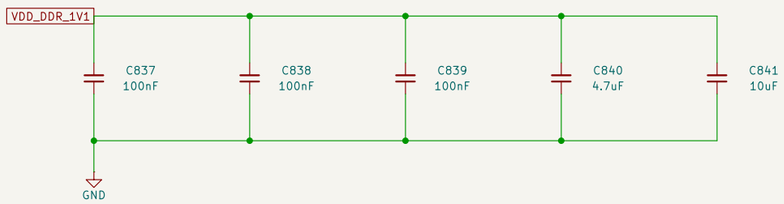
VDD_LOGIC: local bypass; first 4 capacitors under package, remaining capacitors nearby.



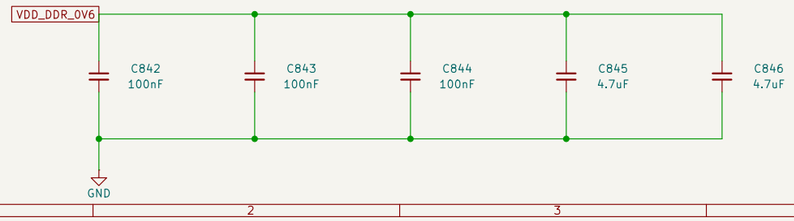
VDD_GPU: local bypass; first 2 capacitors under package, remaining capacitors nearby.



VDD_DDR_1V1: local bypass; first 5 capacitors under package, remaining capacitors nearby.



VDD_DDR_0V6: local bypass; first 5 capacitors under package, remaining capacitors nearby.

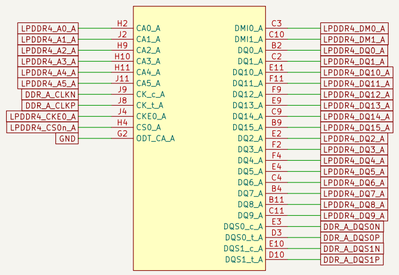


Sheet: /Core local bypass/
File: core-power.kicad_sch

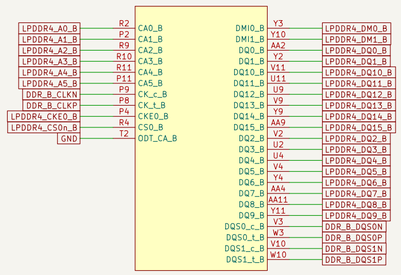
Title: Processor logic, GPU and DDR local bypass – DRAFT

Size: A3	Date: 2026-10-04	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 11/30

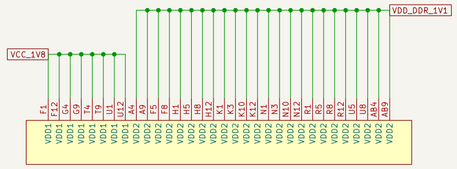
U901A
MT53E512M32D1ZW-046 WT:B



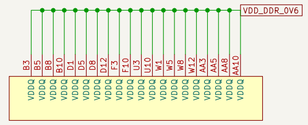
U901B
MT53E512M32D1ZW-046 WT:B



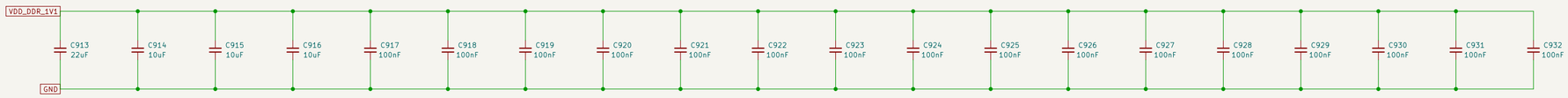
U901C
MT53E512M32D1ZW-046 WT:B



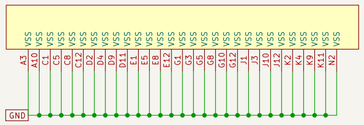
U901D
MT53E512M32D1ZW-046 WT:B



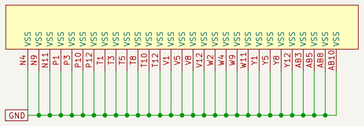
VDD_DDR_1V1: RAM-local bypass; nominal reference values. Physical PDN qualification pending.



U901E
MT53E512M32D1ZW-046 WT:B

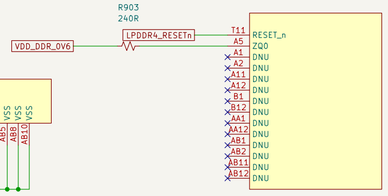


U901F
MT53E512M32D1ZW-046 WT:B

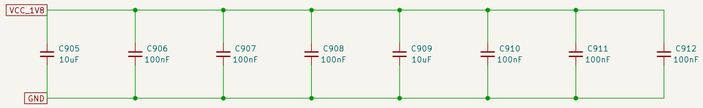


U901G
MT53E512M32D1ZW-046 WT:B

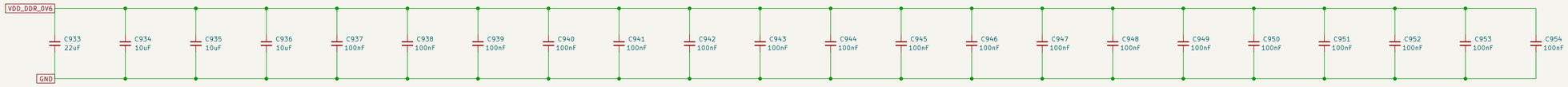
R903
240R



VCC_1V8: RAM-local bypass; nominal reference values. Physical PDN qualification pending.



VDD_DDR_0V6: RAM-local bypass; nominal reference values. Physical PDN qualification pending.

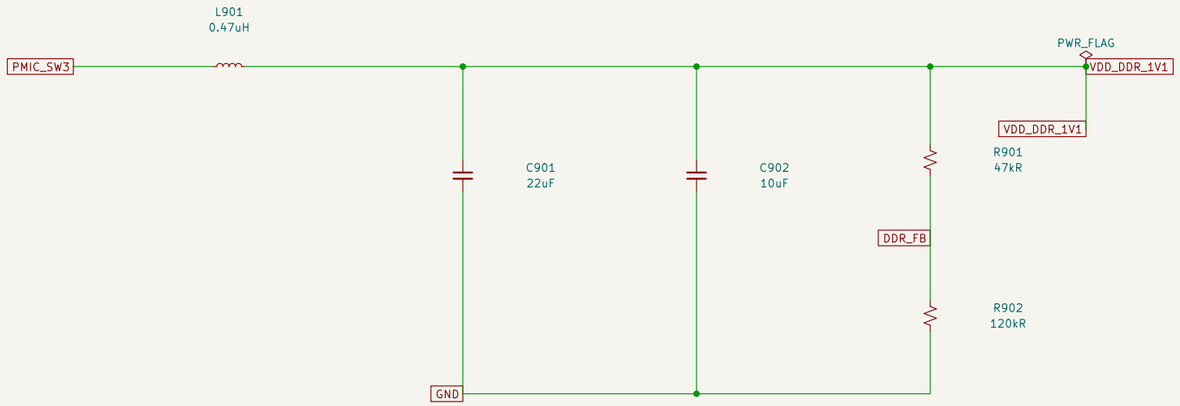


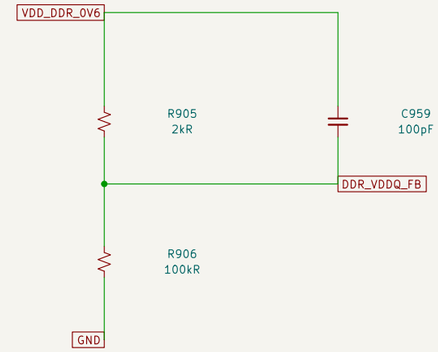
Sheet: /Micron RAM/
File: ram.kicad_sch

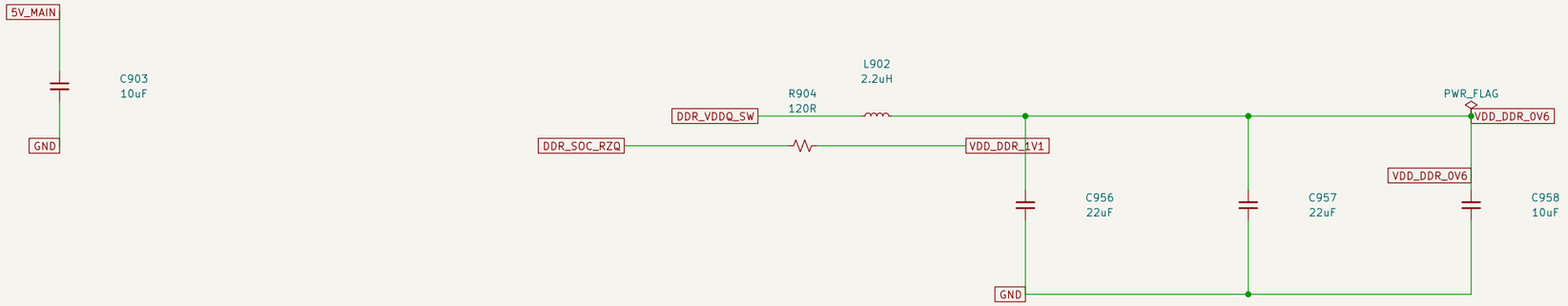
Title: Micron RAM — DRAFT: supplies and DDR bus incomplete

Size: A2 | Date: 2026-10-04 | **Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5 | Id: 12/30

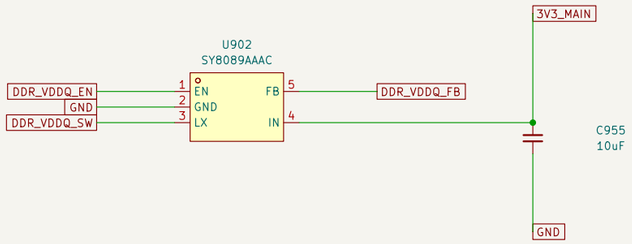
BUCK3: 1.5A rated, external FB reset mode, startup slot 3.
Nominal 1.113V; ripple, load, thermal and shutdown qualification pending.

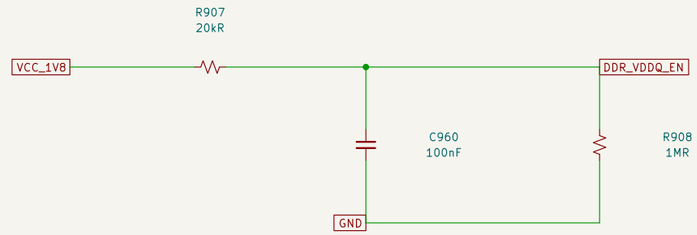






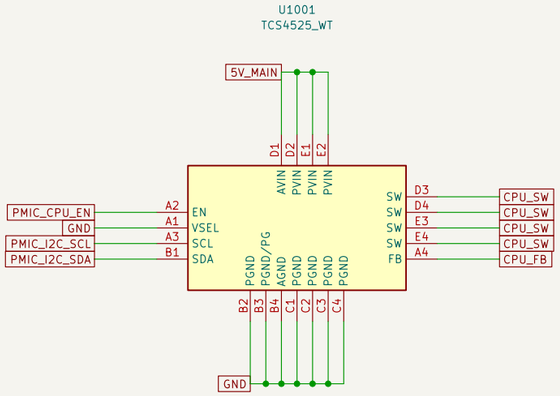
LPDDR4X 0.612V nominal: 2.2uH, 22uF + 22uF + 10uF output.
Place FB sense at load; verify load, ripple, loop and rail order on hardware.



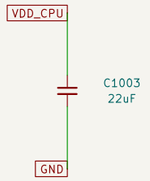


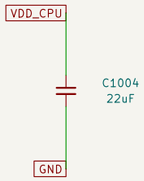
Sheet: /DDR BUCK3/		
File: ddr-power.kicad_sch		
Title: DDR and VDDQ supplies — engineering capture		
Size: A3	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 13/30

CPU DYNAMIC VOLTAGE SCALING

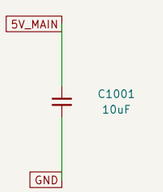


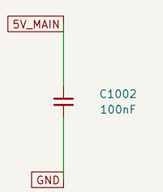


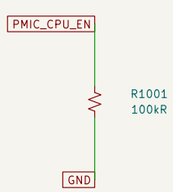


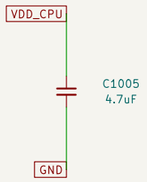


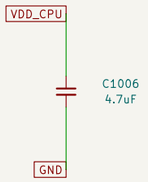


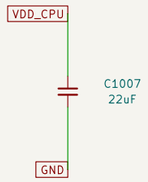


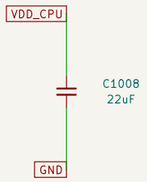


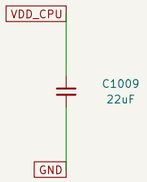












VSEL strapped low: 1.025V reset value: I2C address 0x1C.
RK809 EXT_EN controls startup. Retain DVS via fan53555 driver.
Five local bypass capacitors follow NanoPi R3S power sheet.
Route FB separately to processor load; CPU worst-case load and current revision require closure.

Sheet: /CPU dynamic voltage supply/		
File: cpu-power.kicad_sch		
Title: CPU supply — reference DVS implementation		
Size: A3	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 14/30

16 GB eMMC — 1.8V I/O, HS200 HOST

U1101A
IS21TF16G-JCLI

3V3_MAIN
VCC_1V8
E6 F5 J10 K9 C6 M4 N4 P3 P5
VCC VCC VCC VCC VCCQ VCCQ VCCQ VCCQ VCCQ

EMMC_CLK → M6 CLK
EMMC_CMD → M5 CMD
EMMC_RESET_N → K5 RST_n

DAT0 A3 → EMMC_D0
DAT1 A4 → EMMC_D1
DAT2 A5 → EMMC_D2
DAT3 B2 → EMMC_D3
DAT4 B3 → EMMC_D4
DAT5 B4 → EMMC_D5
DAT6 B5 → EMMC_D6
DAT7 B6 → EMMC_D7
DS H5 → EMMC_STROBE
VDDI C2 → EMMC_VDDI

A6 E7 G5 H10 J5 K8 C4 N2 N5 P4 P6
VSS VSS VSS VSS VSS VSS VSSQ VSSQ VSSQ VSSQ VSSQ
GND

R1103
22R
EMMC_CLK_SOC — EMMC_CLK

3V3_MAIN → C1101 10uF → GND
VCC_1V8 → C1102 1uF → GND
VCC_1V8 → C1103 100nF → GND
EMMC_VDDI → C1104 4.7uF → GND

VCC_1V8 → R1101 10kR → EMMC_CMD
VCC_1V8 → R1102 10kR → EMMC_D0
EMMC_STROBE → R1104 100kR → GND

24 MHz TOTAL ±20 ppm CLOCK

Y1101
SiT8008BI-21-18E-24.000000
VCCA1V8_PMU → 4 VDD
OSC_ENABLE → 1 OE
3 OUT → OSC24_OUT
2 GND → GND

R1105
10kR
VCCA1V8_PMU — OSC_ENABLE

VCCA1V8_PMU → C1105 100nF → GND

R1106
22R
OSC24_OUT — CLK24_IN

U1101B
IS21TF16G-JCLI

A1 A2 A8 A9 A10 A11 A12 A13 A14 B1 B7 B8 B9 B10 B11 B12 B13 B14 C1 C3 C5 C7 C8 C9 C10 C11 C12 C13 C14 D1 → NC
D2 D3 D4 D12 D13 D14 E1 E2 E3 E12 E13 E14 F1 F2 F3 F12 F13 F14 G1 G2 G12 G13 G14 H1 H2 H3 H12 H13 H14 J1 → NC

U1101C
IS21TF16G-JCLI

J2 J3 J12 J13 J14 K1 K2 K3 K12 K13 K14 L1 L2 L3 L12 L13 L14 M1 M2 M3 M7 M8 M9 M10 M11 M12 M13 M14 N1 N3 → NC
N6 N7 N8 N9 N10 N11 N12 N13 N14 P1 P2 P8 P9 P11 P12 P13 P14 A7 E5 E8 E9 E10 F10 G3 G10 K6 K7 K10 P7 P10 → NC
TP1103 GND → NC
R1 R11 → RFU

TP1101
EMMC_CLK
EMMC_CLK

TP1102
EMMC_D0
EMMC_D0

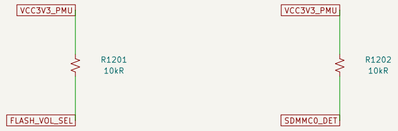
NC/RFU balls remain floating per ISSI page 13.
Use native TLC capacity; measure A/B image and logging endurance.
Keep CLK/D0 MaskROM test branches short.
USB recovery remains independent of eMMC contents.

Sheet: /Boot storage and clock/
File: boot.kicad_sch

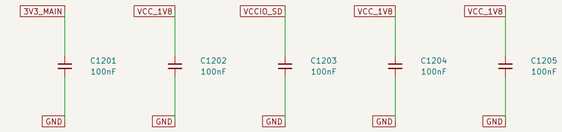
Title: Boot storage and system clock

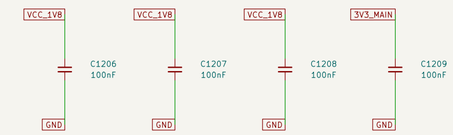
Size: A3	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 15/30

BOOT CONFIGURATION AND LOCAL I/O BYPASS



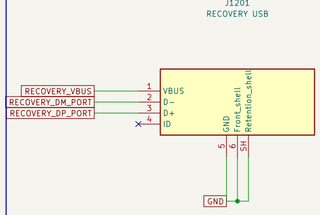
FLASH_VOL_SEL high selects 1.8V eMMC I/O.
SDMMC0_DET high selects debug mux; no SD boot connector fitted.

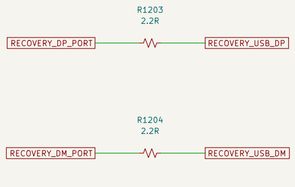


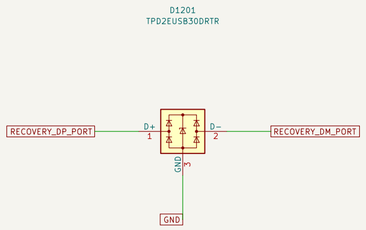


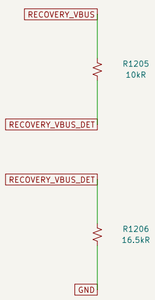
One 100nF capacitor at every VCCIO ball:
1D13, 1C13, 1F17, 1E16, 1N5/6, 1L4/5, 1N8.
Keep unused peripherals disabled. Check domain voltages in DT.

USB OTG0 — DEVICE-ONLY RECOVERY PORT



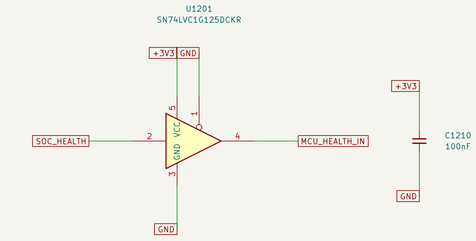


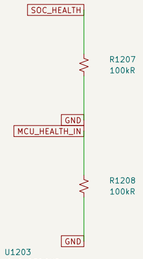


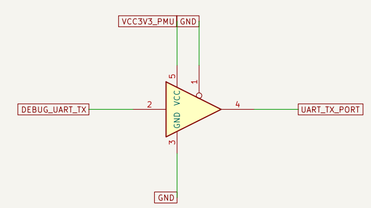


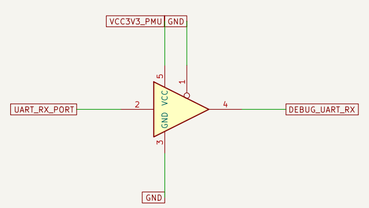
VBUS senses cable presence only: it does not power the board.
10k/16.5k divider: 2.718-3.293V at 4.4-5.25V, 1% resistor corners.
USB OTG ID floats: internal 1.8V pull-up selects device mode.
Place ESD at port: route DP/DM as a short 90-ohm pair.
Recovery through Rockusb/MaskROM still requires device testing.

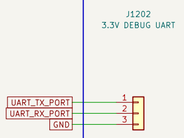
POWERED-OFF-SAFE HEALTH AND DEBUG UART

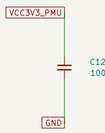


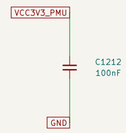








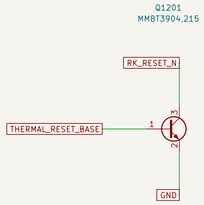


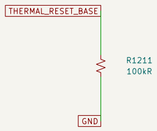




Configure TSADC_SHUT_M0 (1R20) active high.
Hardware overtemperature asserts RK_RESET_N via Q1201.
CPU temperature thresholds and restart policy require qualification.







Sheet: /Host recovery interfaces/
File: host-interface.kicad_sch

Title: Processor recovery and supervision

Size: A2	Date: 2026-10-07	Rev:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 16/30

FB1401
120R @ 100MHz
VDDA_0V9
MULTI_AVDD_0V9

C1401
1uF
MULTI_AVDD_0V9
GND

C1402
100nF
MULTI_AVDD_0V9
GND

C1403
1uF
MULTI_AVDD_0V9
GND

PWR_FLAG
MULTI_AVDD_0V9

C1420
100nF
VDDA_0V9
GND

C1421
100nF
VCCA_1V8
GND

C1422
100nF
VDDA_0V9
GND

C1423
100nF
VCCA_1V8
GND

C1424
100nF
VDDA_0V9
GND

FB1402
120R @ 100MHz
VCCA_1V8
MULTI_AVDD_1V8

C1404
1uF
MULTI_AVDD_1V8
GND

C1405
100nF
MULTI_AVDD_1V8
GND

C1406
1uF
MULTI_AVDD_1V8
GND

PWR_FLAG
MULTI_AVDD_1V8

C1425
100nF
VCCA_1V8
GND

C1426
100nF
VDDA_0V9
GND

C1427
100nF
VCCA_1V8
GND

C1428
100nF
VDDA_0V9
GND

C1429
100nF
VCCA_1V8
GND

FB1403
120R @ 100MHz
VDDA_0V9
USB2_AVDD_0V9

C1407
1uF
USB2_AVDD_0V9
GND

C1408
100nF
USB2_AVDD_0V9
GND

C1409
1uF
USB2_AVDD_0V9
GND

PWR_FLAG
USB2_AVDD_0V9

C1430
100nF
VDDA_0V9
GND

C1431
100nF
VCCA_1V8
GND

C1432
100nF
VDDA_0V9
GND

C1433
100nF
VCCA_1V8
GND

C1434
100nF
3V3_MAIN
GND

Local bypass at powered inactive MIPI/HDMI/eDP analog domains.
Firmware keeps unused PHYs off; check combined RK809 LDO loading.

FB1404
120R @ 100MHz
VCCA_1V8
USB2_AVDD_1V8

C1410
1uF
USB2_AVDD_1V8
GND

C1411
100nF
USB2_AVDD_1V8
GND

C1412
1uF
USB2_AVDD_1V8
GND

PWR_FLAG
USB2_AVDD_1V8

100R || 1uF at DC/DC: feedback remains controlled if Kelvin trace opens.
R1411-R1414 connect FB nets to power at processor load pads.
NPU uses existing R512 (100R). Route FB separately to load.

R1401
100R
VDD_CPU
CPU_FB

R1402
100R
VDD_LOGIC
LOGIC_FB

R1403
100R
VDD_GPU
GPU_FB

MULTI-PHY supplies serve PCIe plus USB3 OTG1.
USB2 filters serve unused host PHY; keep host2/3 disabled.
DCR drop, PDN impedance and reference tolerances require review.

C1441
1uF
VDD_CPU
CPU_FB

C1442
1uF
VDD_LOGIC
LOGIC_FB

C1443
1uF
VDD_GPU
GPU_FB

C1444
1uF
VDD_NPU
NPU_FB

PWR_FLAG
USB_AVDD_0V9

PWR_FLAG
USB_AVDD_1V8

R1411
0R
VDD_CPU
CPU_FB

R1412
0R
VDD_LOGIC
LOGIC_FB

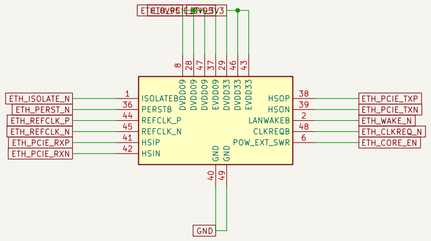
R1413
0R
VDD_GPU
GPU_FB

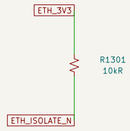
R1414
0R
VDD_NPU
NPU_FB

Sheet: /Peripheral supplies and feedback/
File: peripheral-power.kicad_sch

Title: Processor peripheral supplies and remote feedback

Size: A2 | Date: 2026-10-07 | **Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5 | Id: 17/30

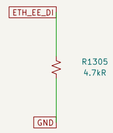




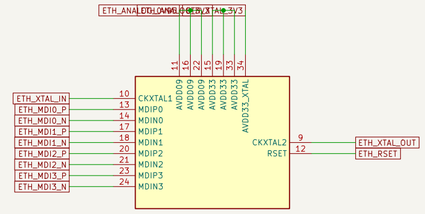




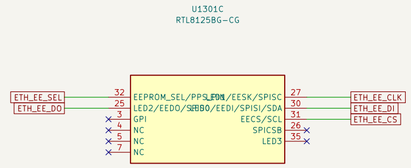


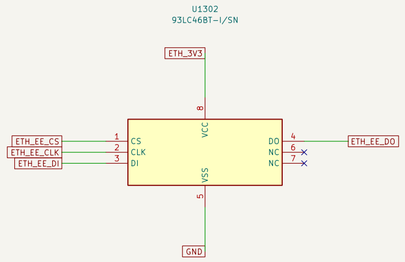






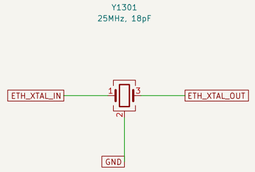


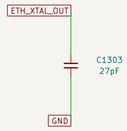










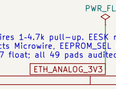


















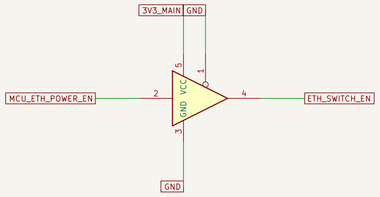
LANWAKE requires 1-4.7k pull-up. EESK requires 4.7k pull-up; low enters test mode.
EEDI low selects Microwire. EEPROM_SEL low selects 93C46. All strap levels at power-on.
BG pins 4/5/7 float; all 49 pads audited. 2.5G magnetics and four-pair PoE attached; qualify hardware.

Sheet: /2.5GbE controller/
File: nic.kicad_sch

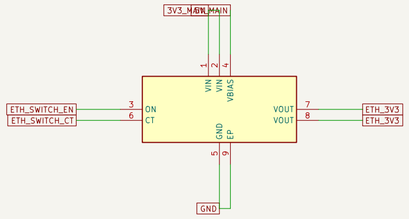
Title: 2.5GbE controller — RTL8125BG external-core variant

Size: A2	Date: 2026-10-07	**Rev:**
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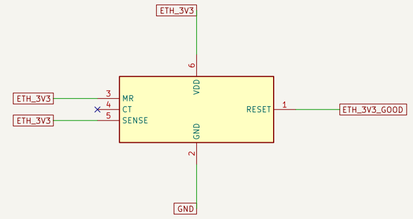
U1501
SN74LVC1G125DCKR

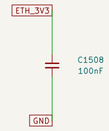


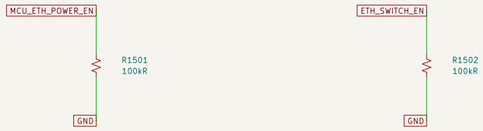
U1502
TPS22965DSGR

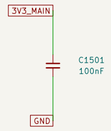


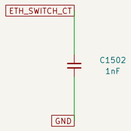
U1504
TPS3808G33DBVR

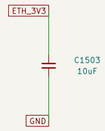


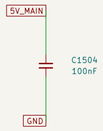














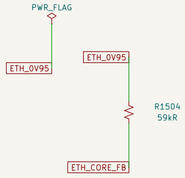
Ioff buffer prevents MCU back-power when main supply is off.
CT=1nF, VBIAS=5V: 1.33ms typical 3.3V rise (TI Table 1).
Quick-output discharge 225R typical: measured decay required.

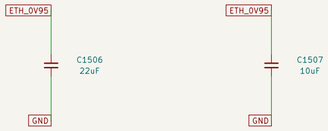
U1503
SY8089AAAC

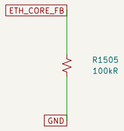


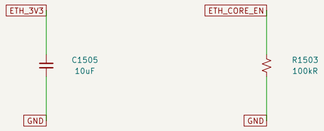
L1501
2.2uH





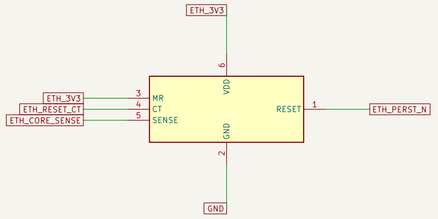




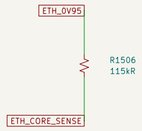


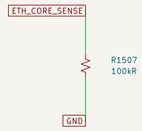
0.954V nominal: .588-.612V FB and 0.1% R: 0.934-0.974V.
RTL8125BG .92-.98V operating rail, 650mA maximum (Table 29).
POW_EXT_SWR controls core startup; do not fit RTL8125BGS here.

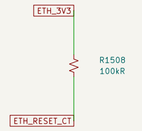
U1505
TPS3808G01DBVR













G33: open CT gives 12-28ms delay to clock-enable logic.
G01: CT via 100k to VDD gives 180-420ms reset delay.
0.871V nominal core supervision; worst release <=0.925V.
This detects collapse; it does not guarantee the .92V operating minimum.
Hold PERST for >=100ms with clock stable. Verify all corners on bench.

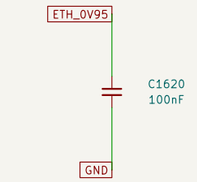
Sheet: /Ethernet power sequencing/
File: nic-power.kicad_sch

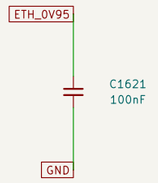
Title: Ethernet power and hardware reset sequencing

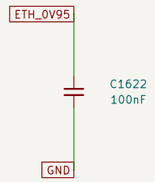
Size: A2	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 19/30

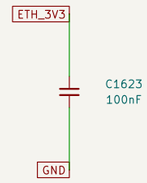
C1601
100nF
PCIE_TXP_SOC
ETH_PCIE_RXP
C1602
100nF
PCIE_TXN_SOC
ETH_PCIE_RXN
C1603
100nF
ETH_PCIE_TXP
PCIE_RXP_SOC
C1604
100nF
ETH_PCIE_TXN
PCIE_RXN_SOC
100-ohm differential pairs: one AC-coupling capacitor per conductor at transmitter.
SoC TX reaches NIC HSIP/HSIN; NIC HSOP/HSON reaches SoC RX.
Keep PCIe controller quiescent before Ethernet power removal.
U1601
SN74LVC1G07DCKR
ETH_3V3
VCC
PCIE_PERST_SOC_N
A
N.C
Y
ETH_PERST_N
GND
C1605
100nF
U1602
SN74LVC1G07DCKR
VCC3V3_PMU
ETH_CLKREQ_N
PCIE_CLKREQ_N
C1606
100nF
R1602
4.7kR
U1603
SN74LVC1G07DCKR
ETH_WAKE_N
PCIE_WAKE_N
C1607
100nF
R1603
4.7kR
U1604
TMUXHS4212IRKSR
PCIE_REFCLK_P
PCIE_REFCLK_N
ETH_CLOCK_OE_N
A0p
A0n
OEn
SEL
A1p
A1n
RSV01
RSV02
B0P
B0N
B1P
B1N
C0P
C0N
C1P
C1N
ETH_REFCLK_P
ETH_REFCLK_N
RESET sinks: SoC U1601, supervisor Q303, core monitor U1505.
No push-pull outputs share ETH_PERST_N. CLKREQ and wake outputs use ioff buffers.
While NIC is off, firmware ignores asserted wake/CLKREQ and disables ASPM.
C1608
100nF
R1604
10kR
ETH_3V3_GOOD
ETH_CLOCK_BASE
Q1601
MMBT3904,215
R1605
100kR
R1606
10kR
Clock disconnected by default. Valid switched 3.3V after U1504 delay turns Q1601 on.
Mux stays on the SoC supply; NIC power cycling cannot turn on its clock path.
Core reset delay ensures >100us clock-before-PERST margin. Validate shutdown transients.
Sheet: /PCIe and power-cycle interfaces/
File: nic-interface.kicad_sch
Title: PCIe and powered-off-safe Ethernet interfaces
Size: A2
Date: 2026-10-07
Rev:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5
Id: 20/30

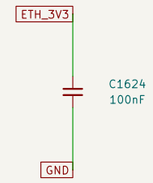
One 100nF at each of 14 RTL8125BG supply pins.
Place bulk beside filtered analog banks; check DC drop and impedance under maximum load.

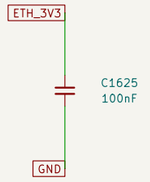


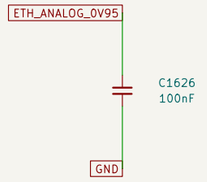


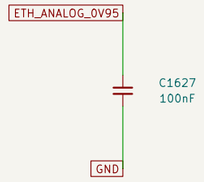


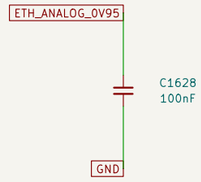


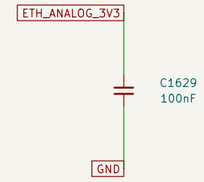


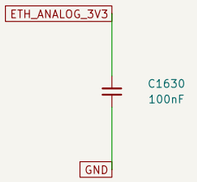


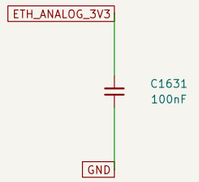


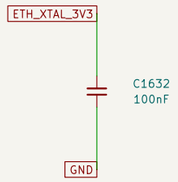


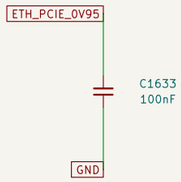


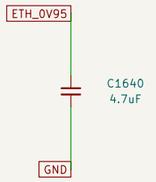


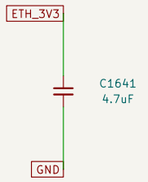


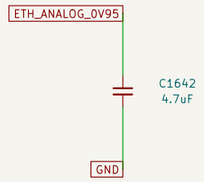


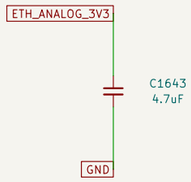


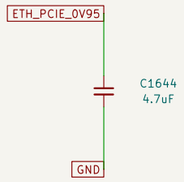










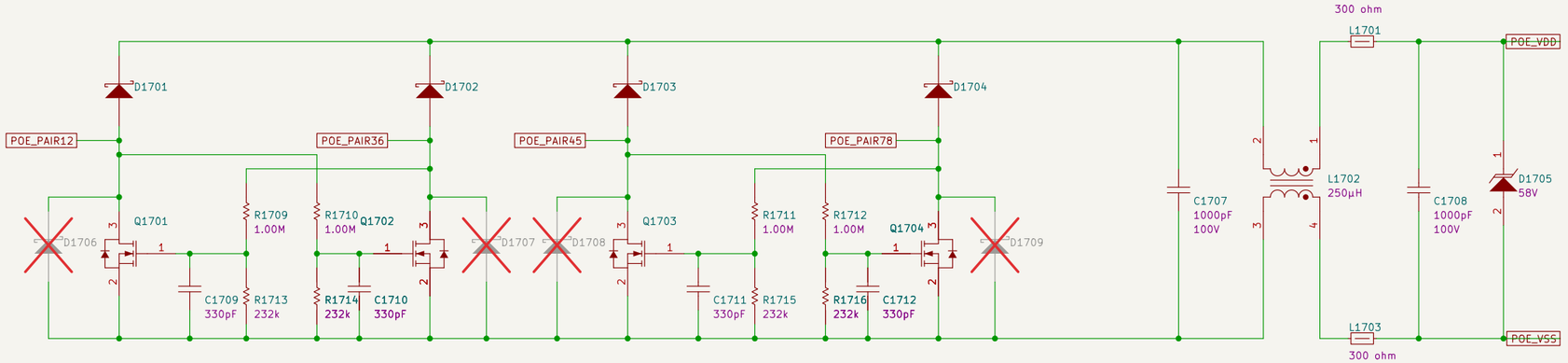


Sheet: /Ethernet local bypass/
File: nic-bypass.kicad_sch

Title: Ethernet local supply bypass

Size: A3	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 21/30

Cable pair feeds from HX4110NL 2.5G / four-pair 60W ingress.
Reference active bridge retained. Permanent VSS1-to-VSS current-link is now a PCB trace.
POE_VSS and POE_RTN remain isolated from system GND.



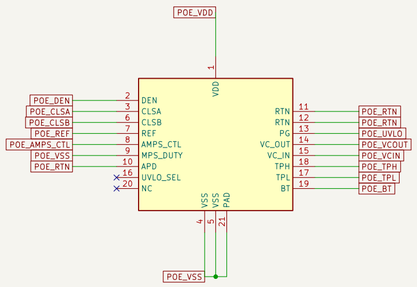
Sheet: /PoE active input bridge/
File: poe-input.kicad_sch

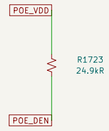
Title: Isolated PoE — input, TI TIDA-050023 derivative

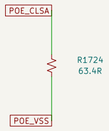
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TID #:	Date: 2026-10-07	
Number:	KiCad E.D.A. 10.99.0-rc1-dirty-g83b5faf3d5	Rev:
SVN Rev:	Assembly Variant:	Sheet: of
Drawn By:	File:	Size: B
Engineer:	Contact:	http://www.ti.com

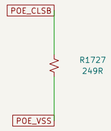
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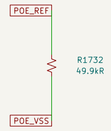
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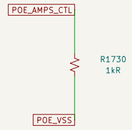




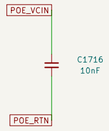


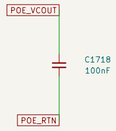


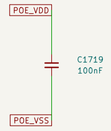
















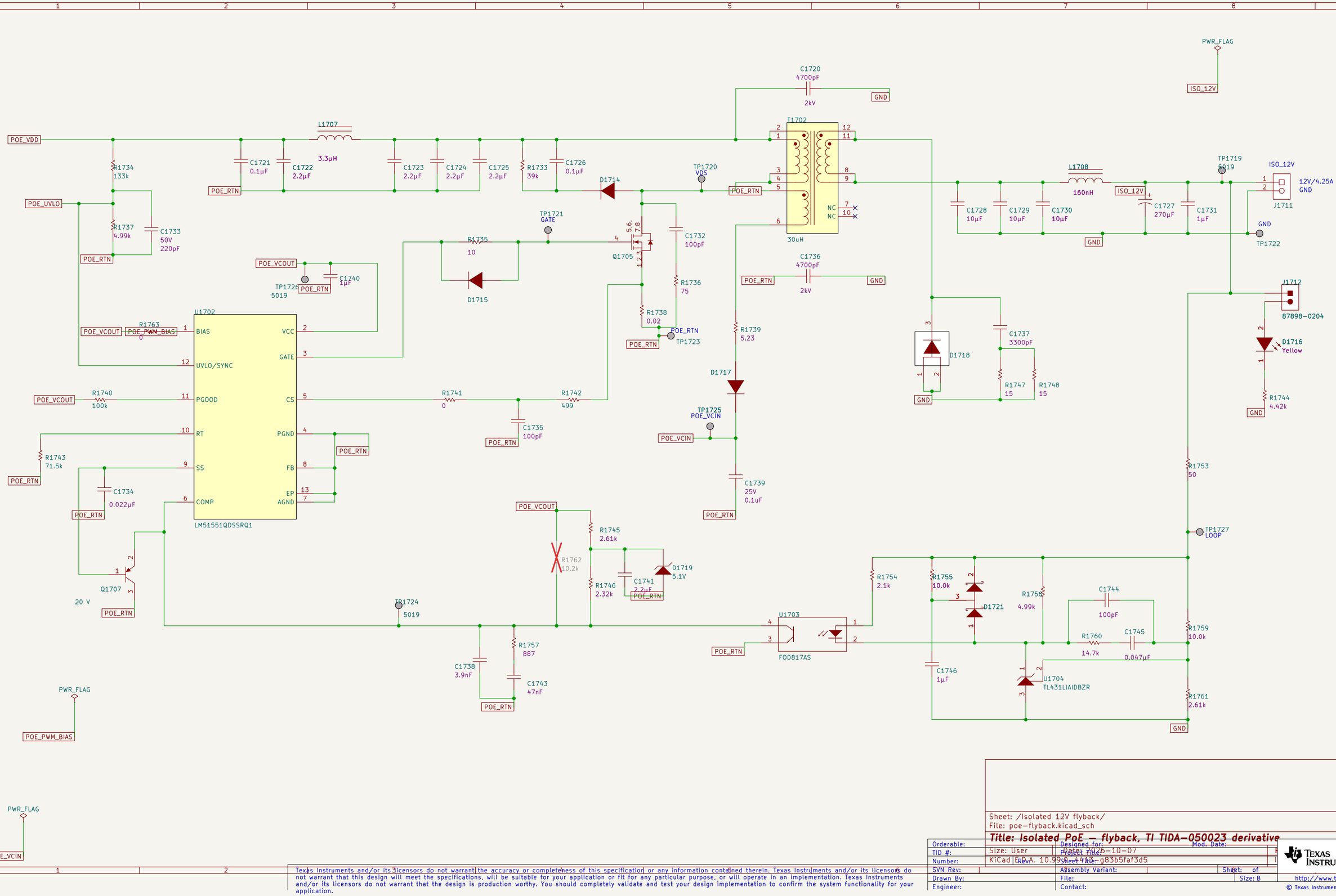


Fixed Class 6: CLSA 63.4R, CLSB 249R. Accept only 51W bt grant: TPH low / TPL high / BT low.
Three optocouplers carry status across isolation to the always-on supervisor.
No auxiliary input. APD tied to RTN: detection resistor changes from reference 27.4k to 24.9k
because its auxiliary-divider parallel load was removed. MPS_DUTY tied to VSS (12.5% bt).
UVLO_SEL left open, following the LM51551 TI reference: validate startup at input corners.
POE_VSS is the cable return. POE_RTN is the switched primary return. Neither connects to GND.

Sheet: /PoE PD / Class 6/
File: poe-pd.kicad_sch

Title: Class-6 powered device — fixed assembly configuration

Size: A2	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 23/30



Sheet: /Isolated 12V flyback/
File: poe-flyback.kicad_sch

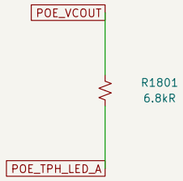
Title: Isolated PoE – flyback, TI TIDA-050023 derivative

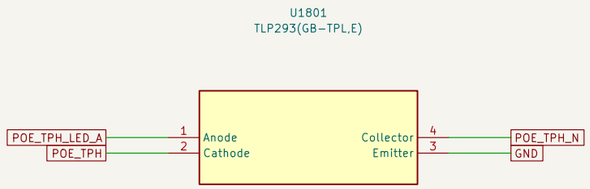
Orderable:		Designed for:	Mod. Date:
TID #:		Size: User	Date: 2026-10-07
Number:		KiCad EDA 10.99.0-rc2-g83b5faf3d5	Rev:
SVN Rev:		Assembly Variant:	Sheet: of
Drawn By:		File:	Size: B
Engineer:		Contact:	

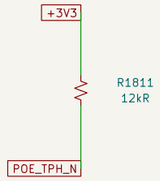
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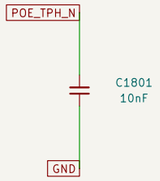
http://www.ti.com

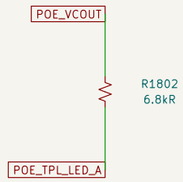
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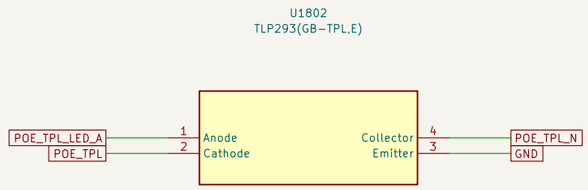


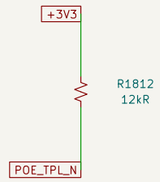


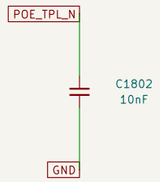


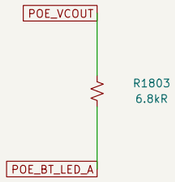


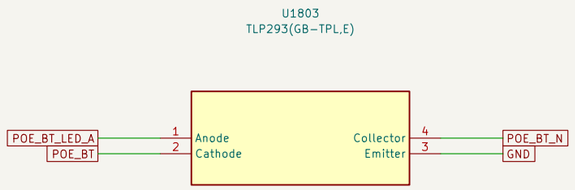


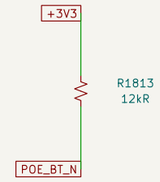


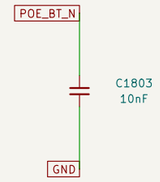












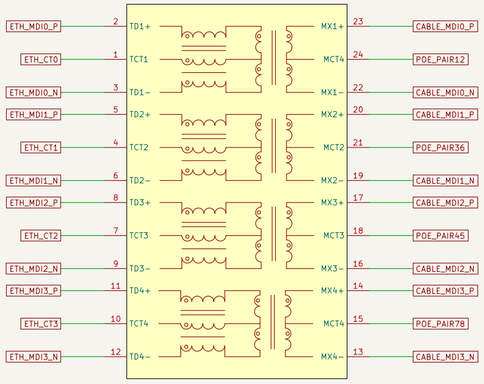
Primary LEDs are driven by TPS2373 open-drain outputs referenced to POE_RTN.
Secondary outputs: PB10 / PB11 / PB12 (MCU pads 22 / 23 / 24), active low.
Class 6 grant: TPH_N=0, TPL_N=1, BT_N=0, stable after startup.
Reject all other combinations. Keep MAIN5V, modem and Ethernet power off until grant.
6.8k limits each LED to <2.3mA at 15V; TPS2373 recommended sink <=3mA.
12k collector pull-up and 10nF filter. Verify CTR/leakage at -40 to 85C and bias corners.
All-high also means unpowered / invalid status. Loss of grant cuts main loads.
Do not join POE_RTN, POE_VSS or POE_VDD to secondary GND.

Sheet: /Isolated PoE allocation/
File: poe-status.kicad_sch

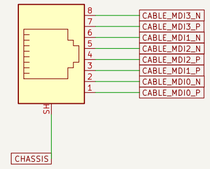
Title: Isolated PoE allocation — supervisor inputs

Size: A3	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 25/30

T1701
HX4110NLT

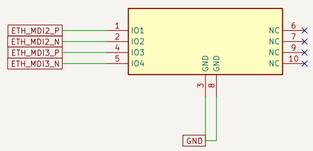


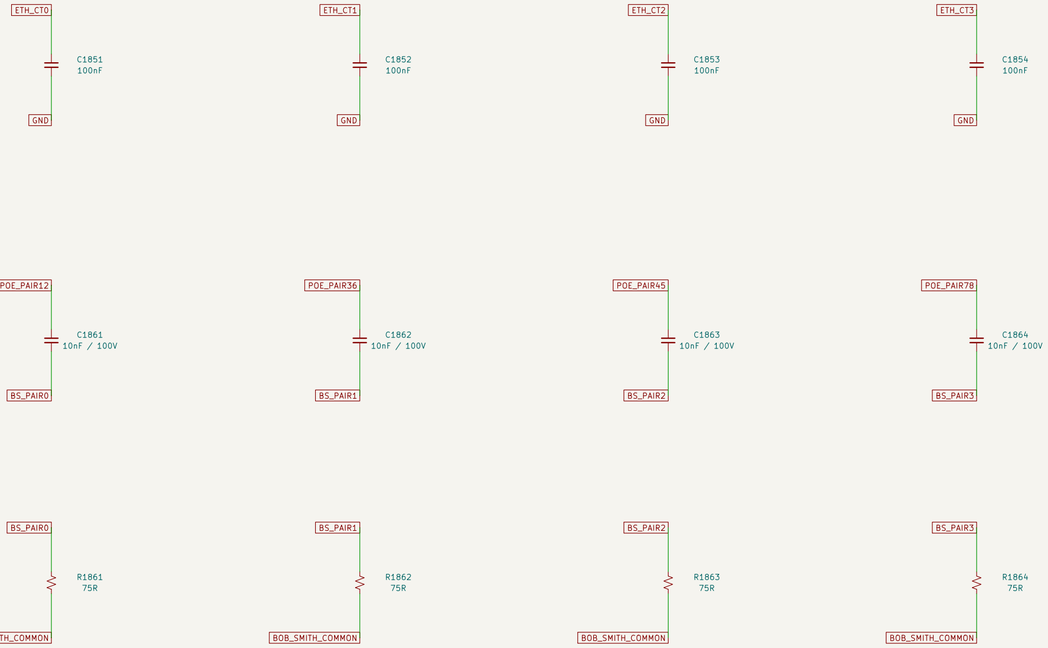
J1701
2.5GbE + PoE

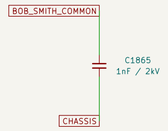


D1851
TPD4E02B04DQAR

D1852
TPD4E02B04DQAR







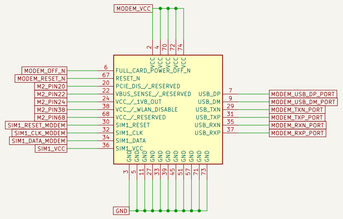
T1701 pin/polarity and four center taps audited to Pulse page 2.
PHY CTs: one 100nF to GND each, following RTL8125BG reference.
Cable CTs feed active PoE bridge: 10nF series caps block PoE DC in Bob-Smith termination. CHASSIS is RJ45 shell/enclosure only.
No copper connection across magnetics or to primary PoE returns.
Exact transformer and RJ45 footprints remain a layout gate.

Sheet: /Ethernet cable ingress/
File: ethernet-ingress.kicad_sch

Title: 2.5GbE cable ingress and four-pair PoE extraction

Size: A2	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 26/30

U1901
TPS54620RHLR
ISO_12V
MODEM_BUCK_EN
MODEM_BUCK_RT
MODEM_BUCK_FB
MODEM_BUCK_COMP
MODEM_BUCK_SS
MODEM_BUCK_BOOT
MODEM_BUCK_SW
MODEM_BUCK_GOOD
GND
C1901
100nF
R1901
36.3kR
R1902
10kR
L1901
3.3uH
C1902
22uF
C1903
22uF
C1904
22uF
C1905
10uF
C1906
10uF
C1907
100nF
U1902
TPS25947ALRPWR
MODEM_FUSE_EN
MODEM_FUSE_OVLO
MODEM_FUSE_PGTH
MODEM_FUSE_ILM
MODEM_FUSE_DVDT
MODEM_VCC
MODEM_SWITCH_GOOD
R1913
237kR
R1915
165kR
R1914
100kR
R1916
100kR
R1903
100kR
R1904
10kR
R1905
100kR
R1906
10kR
R1911
10kR
R1912
100kR
R1918
10kR
R1917
562R
C1911
10nF
C1912
100nF
C1908
47nF
R1907
1kR
C1909
100nF
C1910
1nF
PWR_FLAG
+3V3
MODEM_3V7_RAW
MODEM_COMP_RC
Compensated current-mode regulator supports the downstream 220uF modem bulk bank.
Type II: 1k / 100nF, 1nF high-frequency pole. Include eFuse resistance and both banks
in loop analysis. Verify loop, burst droop, soft-start and thermal behavior on hardware.
5A is a provisional burst design allocation. The selected 1A/power-contact M.2 socket permits
5A total at 25C; sharing, hot derating and module envelope require qualification.
This is not a guaranteed current envelope for RM551E-GL or Sierra mmWave accessories.
Latch-off eFuse: 45mOhm maximum RON. 5629/0.1% gives approx. 5.27-6.45A threshold
using the 5498 guaranteed table, NOT 5A continuous connector qualification. ITIMER open.
OVLO: 237k/100k. 3.93-4.18V resistor/reference corners, below modem 4.4V limit.
PGTH: 165k/100k, nominal 3.18V. Use separate always-on output monitor for USB isolation.
Maximum continuous device current is 5.5A; connector/profile hot limits can be lower.
No UL assembly claim; this threshold does not meet the datasheet RILIM>=750R recognition condition.
U1903
TPS380BG01DBVR
MODEM_RAIL_GOOD
MODEM_MON_CT
MODEM_MON_SENSE
R1921
330kR
R1922
47kR
C1913
100nF
U1904
SN74LVC1G04DCKR
MCU_MODEM_POWER_EN
MODEM_DISCHARGE_GATE
R1923
10kR
C1914
100nF
Q1901
AO3400A
MODEM_DISCHARGE_DRAIN
MODEM_DISCHARGE_GATE
R1924
100R / 0.25W
R1925
100kR
R1926
100kR
U1903 is powered from always-on 3.3V; output threshold nominal 3.249V.
CT via 100k to VDD; 180-420ms delay exceeds Sierra 150ms no-drive interval.
MODEM_RAIL_GOOD drives data isolation and MCU PB13 (pad25). MCU PA0 controls power.
Low command enables Q1901 discharge; wait at least 1s and bench-verify socket <0.1V
before restart. MCU default/high-Z command is off. Qualify timing per modem profile.
Clear a latched fault only after determining cause; do not create an autonomous reset loop.
Sheet: /Modem independent power/
File: modem-power.kicad_sch
Title: Independent modem power — 3.7V / burst allocation 5A
Size: A1
Date: 2026-10-07
Rev:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5
Id: 27/30



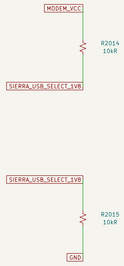


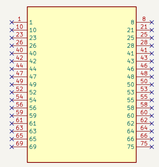




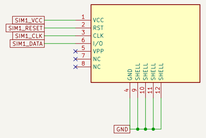












Default assembly: R2011/12/13/16/17 DNP (Quectel). NEVER feed Quectel pins 24,38,68.
Sierra USB profile: fit all five links; pins 24/38/68 add VCC, 20 selects USB, 22 detects VBUS.
Pin 20 divider is powered by modem VCC: include module pulldown in level verification.
RESET uses module internal 1.8V pull-up; Q301/Q302 only sink reset/power-off signals.
All unused PCIe, coexistence, SIM2, antenna control and configuration contacts float.
No host drive before valid rail delay. No mmWave antenna power hardware is captured.
Every profile remains on release hold pending current controlled module guide and power envelope.



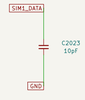




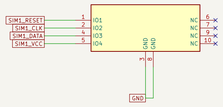






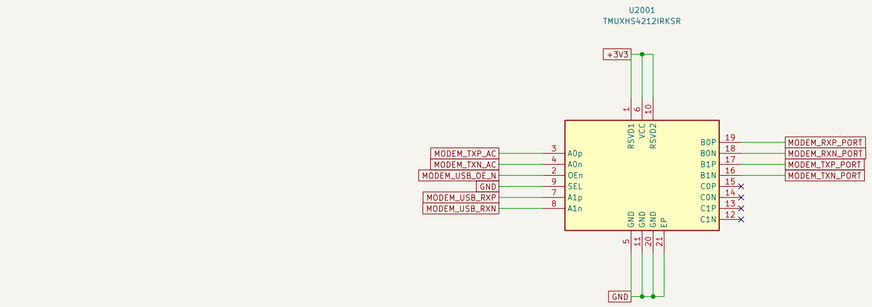




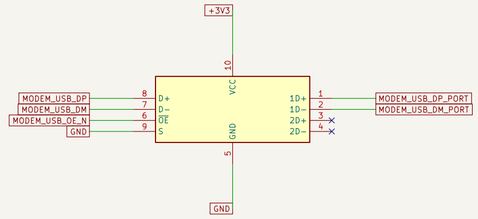


SIM routing <=200mm and away from RF/SW. 22R/10pF and 15k follow RM520 reference.
Disable AT+QSIMDET; keep modem SIM-detect contact 66 open. Sierra open means card present.
Validate SIM ATR/retry behavior with card present/absent and each profile.

Sheet: /Modem and SIM sockets/		
File: modem-socket.kicad_sch		
Title: Modem socket and SIM — assembly profiles required		
Size: A1	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 28/30





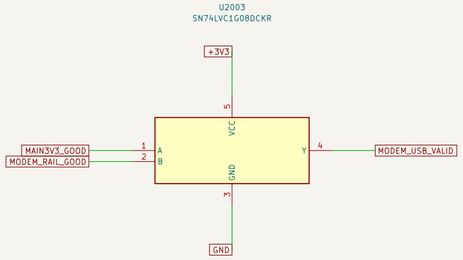




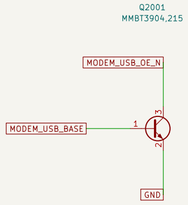


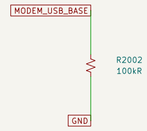


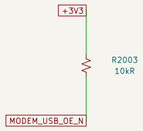












Both USB muxes remain powered when modem/SoC are off. Hardware AND requires valid host and delayed modem rails before enabling data. U1903 releases after 180-420ms of valid VCC. Host TX -> module RX (37P/35N). Module TX (31P/29N) -> host RX. Module contains TX capacitors. No M.2 PCIe routing; dedicated SoC PCIe port serves Ethernet. USB3 eye/jitter/loss and rail-fall isolation timing require measurement; capture/ERC does not establish recovery or throughput.

Sheet: /Modem USB isolation/
File: modem-interface.kicad_sch

Title: Modem USB data isolation and coupling

Size: A2	Date: 2026-10-07	Rev:
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 29/30

C2030
220uF / 6.3V

MODEM_VCC

+

GND

M2_PIN24

C2036
100nF

GND

M2_PIN38

C2037
100nF

GND

M2_PIN68

C2038
100nF

GND

MODEM_VCC

C2031
100nF

GND

MODEM_VCC

C2032
100nF

GND

MODEM_VCC

C2033
100nF

GND

MODEM_VCC

C2034
100nF

GND

MODEM_VCC

C2035
100nF

GND

MODEM_VCC

C2041
220pF

GND

MODEM_VCC

C2042
68pF

GND

MODEM_VCC

C2043
15pF

GND

MODEM_VCC

C2044
9.1pF

GND

MODEM_VCC

C2045
4.7pF

GND

C2030 and RF bank at socket. One 100nF at each of five common VCC contacts.
C2036/37/38 SIERRA ONLY: DNP in Quectel profiles along with their feed links.
Confirm effective MLCC capacitance at DC bias and full modem internal capacitance.
Layout must share current across every populated VCC/GND contact and minimize drop.
No released four-module burst specification or temperature/thermal qualification yet.

Sheet: /Modem local bypass/
File: modem-bypass.kicad_sch

Title: Modem local bulk and RF bypass

Size: A3	Date: 2026-10-07	**Rev:**
KiCad E.D.A. 10.99.0-4413-g83b5faf3d5		Id: 30/30